



BALAMURUGAN C

AI & Data Science Engineer | Security Researcher

tharanishbalaa@gmail.com +91 9159850002 Coimbatore, Tamil Nadu github.com/Balamurugan0113
linkedin.com/in/balamurugan-c-5507b82a3

PROFESSIONAL SUMMARY

AI & Data Science Engineer with expertise in machine learning pipelines, penetration testing, and network security. Final-year B.Tech student with 5+ AI models developed, 20+ security lab audits, and 10+ freelance projects delivered. Skilled in Python, TypeScript, cloud infrastructure, and security tooling. Published researcher in IEEE conferences on AI-driven security analytics.

EXPERIENCE

Security Research Intern Jun 2024 - Present
Cybersecurity Research Lab

- Advanced research on ML-based network intrusion detection systems.
- Developed novel ML-based NIDS achieving 99.2% detection rate with <0.1% false positives
 - Published 2 research papers in IEEE conferences on AI-driven security analytics
 - Contributed 500+ commits to open-source security tools (Nmap scripts, Metasploit modules)
 - Led weekly security workshops training 50+ students on penetration testing methodologies

Full-Stack Developer & Security Consultant Jan 2023 - Present
Freelance / Self-Employed

- Enterprise web applications and security assessments for 10+ clients.
- Delivered 10+ production client projects with 100% satisfaction rate
 - Conducted 50+ security audits identifying 200+ vulnerabilities
 - Built reusable component library reducing development time by 60%
 - Maintained 99.9% SLA uptime across all client deployments

AI/ML Open Source Contributor 2022 - Present
Various Projects

- Core contributor to security-focused ML library with 2K+ GitHub stars
- Implemented GPU-optimized inference pipeline reducing latency by 3x for ONNX Runtime
- Created tutorials used by 10K+ developers for ML security applications

KEY PROJECTS

Real-Time Network Intrusion Detection System
Multi-threaded packet capture and analysis system decoding raw IP/TCP headers to detect port scans, signature anomalies, and cleartext credential transmission at 10Gbps+ line rate.
Python, Network Security, Packet Analysis, TimescaleDB

Real-Time Face Recognition Attendance System
Two-stage pipeline: Haar Cascade face detection + dlib ResNet embedding extraction achieving 99.4% recognition accuracy on live webcam feeds.
OpenCV, Python, Face Recognition, SQLite, Streamlit

Campus Feed Sentiment Analyzer
NLP pipeline processing real-time student forum feedback via WebSockets using DistilBERT with <200ms inference latency to detect negative sentiment spikes.
NLP, FastAPI, WebSockets, Transformers, ONNX Runtime

IoT Real-Time AQI Predictor & Alert System
ESP32 sensor nodes publishing PM2.5/PM10 telemetry via MQTT with Random Forest forecasting achieving 95% prediction correlation.
IoT, ML, Random Forest, MQTT, InfluxDB

Real-Time ASL Hand Sign Translator

SKILLS

Penetration Testing
Network Pentesting, Web App Auditing, API Security Assessment, Vulnerability Assessment

Programming Languages
Python, JavaScript / TypeScript, Bash / PowerShell, SQL

Security Tools
Burp Suite, Nmap, Wireshark, Metasploit Framework

Cloud & Infrastructure
AWS Security, Docker, Linux Hardening, Database Security

AI/ML Frameworks
TensorFlow, PyTorch, Scikit-learn, ONNX Runtime, HuggingFace

PROFICIENCY

Penetration Testing	95%
Security Tools	90%
Programming Languages	88%
Cloud & Infrastructure	82%

CERTIFICATIONS

CEH Master (Certified Ethical Hacker)
EC-Council · 2023

OSCP (Offensive Security Certified Professional)
Offensive Security · 2024

AWS Certified Security - Specialty
Amazon Web Services · 2024

EDUCATION

B.Tech in Artificial Intelligence & Data Science 2021 - 2025
Info Institute of Engineering, Coimbatore
CGPA: 8.9/10 · Top 10% of class

Higher Secondary (HSC) 2021
Shri Nehru Vidyalaya Matric Hr Sec School

ACHIEVEMENTS

- Smart India Hackathon 2025 - Finalist
- Trisquadathon 2024 - Winner
- Led university cyber security club (200+ members)
- Organized 3 national-level CTF competitions (500+ participants)

MediaPipe landmark extraction + Bi-LSTM model processing camera streams at 24 FPS for American Sign Language translation.
[Deep Learning](#), [MediaPipe](#), [LSTM](#), [TensorFlow](#), [TensorFlow.js](#)